

Cointegration-Based Pairs Trading Model

(JBLU/LUV)

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Strategy: Statistical Arbitrage (Mean Reversion) | Sector: Airlines

The Setup

Testing all 15 possible pairs across six major US airlines (DAL, UAL, AAL, LUV, ALK, JBLU) for cointegration using the Engle-Granger test, Southwest (LUV) and JetBlue (JBLU) emerged as the strongest candidate by a wide margin ($p = 0.007$), well below the standard 0.05 significance threshold.

This result was contradictory to the initial eye test, where UAL and ALK seemed to be the most cointegrated pair. However, they tested weakly ($p=0.77$), showing how correlation and cointegration are different: two stocks can visually move together even if their spread does not have stable, mean-reverting behaviour.

Methodology

Using 70% of the time series data on the six tickers (retrieved through the “yfinance” package) from 2019 to mid 2024 as an in-sample strategy building period and using the other 30% as an untouched out-of-sample testing period, a hedge ratio of 1.76 was retrieved through OLS regression. This spread oscillated consistently around a stable mean throughout the entire in-sample period, including during the extreme COVID-19 demand shock in 2020, when it deviated furthest from its average before reverting, supporting the case for a genuine underlying relationship rather than a coincidental one.

A rolling 30-day z-score was constructed from this spread, and a mean-reversion trading rule was applied: enter a position when the z-score exceeds ± 2 (betting on reversion), and exit when it returns to 0. Positions were shifted forward by one day to avoid lookahead bias, ensuring trades only executed on information available at the time.

Results

Metric	Result
Sharpe Ratio	0.27
Max Drawdown	-10.7
Number of Trades	29

The strategy was profitable out-of-sample, on data entirely unseen during the design phase, and transaction costs had minimal impact on returns, consistent with the strategy's relatively low trading frequency (roughly one trade every four weeks). The Sharpe ratio of 0.27 is modest rather than exceptional, and the maximum drawdown of -10.7 represents a significant pullback relative to the strategy's peak cumulative gains, worth weighing seriously against the returns achieved.

Parameter Sensitivity

To test whether this result depended heavily on one specific configuration, several alternative lookback windows and thresholds were tested:

- **5-day lookback, entry $z = 2.0$:** z -score never crossed the threshold, 0 trades made.
- **5-day lookback, entry $z = 1.0$:** lowering the threshold to compensate reactivated trading, but the strategy lost money out-of-sample, likely due to a combination of overtrading on short-term noise and increased transaction cost drag from higher trade frequency.
- **30-day lookback, entry $z = 1.5$:** performance remained stable relative to the primary configuration (Sharpe 0.30 vs. 0.27, drawdown -11 vs. -10.7), with trade count increasing from 29 to 41 as expected from a looser threshold.

This sensitivity analysis suggests the 30-day lookback captures a genuinely more stable signal than the 5-day alternative, and that results are not fragile to small changes in the entry threshold within that window, a modest but meaningful piece of evidence against overfitting to one specific parameter choice.

Limitation and Further Work

With 29-41 trades over roughly 2.5 years of out-of-sample data, the sample size remains modest for making strong statistical claims about a repeatable edge. A more rigorous version of this project would test the same methodology across multiple pairs and sectors, use a longer or multiple out-of-sample windows via rolling walk-forward validation, and incorporate short-borrow costs explicitly rather than only trade-execution costs. The economic rationale for cointegration (shared fuel cost exposure, overlapping leisure-travel customer base between two budget-oriented carriers) is plausible but was not independently verified against fundamental data, and would strengthen the thesis if confirmed.